Vehicle Tracking + Occlusions Research Project

Phase 1G Research Documentation

Trigger-Driven Post-Occlusion Vehicle Candidate Generation

Purpose: archival documentation of the design decisions, implementation steps, experiments, results, corrections, limitations, and frozen configuration produced during Phase 1G.

Phase status:	Complete and frozen
Dataset used:	UA-DETRAC pilot events
Real detector:	YOLO11n (COCO vehicle classes)
Final operating point:	imgsz=640, confidence=0.20
Phase dependency:	Phase 1F persistent CRITICAL / LOST trigger
Next phase:	Phase 1H controlled recovery alternatives

Documentation generated from the final Phase 1G experimental record. Earlier provisional results are retained where they explain a design decision, but the corrected frozen results are explicitly identified.

Document Map

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1 Phase Objective and Research Boundary

Phase 1G was designed to answer a narrow question: *once Phase 1F has determined that the selected target is persistently ambiguous or lost, can the system generate a manageable set of plausible road-vehicle candidates that contains the true target when it becomes visible again?*

This phase deliberately does **not** solve final identity recovery. The user selects the target vehicle once through the interface at the beginning of tracking. After an occlusion, Phase 1G automatically proposes candidate vehicles. A later phase must decide which candidate, if any, is the originally selected target.

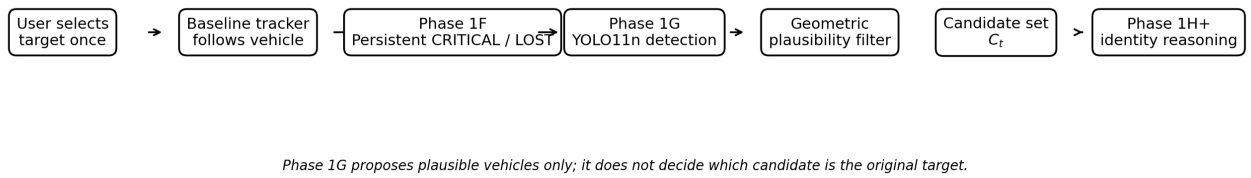


Figure 1: Position of Phase 1G in the recovery pipeline. The user performs the initial target selection only; post-occlusion candidate generation and later identity reasoning are automatic.

The conceptual boundary is therefore:

- **Initial user action:** select one target vehicle in the video.
- **Phase 1F:** determine whether tracking is reliable, persistently critical, or lost.

- **Phase 1G:** detect and filter plausible road-vehicle candidates.
- **Phase 1H and later:** compare identity evidence and produce MATCH / REJECT / NEED MORE EVIDENCE behavior.

2 Trigger Contract from Phase 1F

Phase 1G is activated only when Phase 1F reports either:

1. **LOST**, or
2. **persistent CRITICAL**.

A transient CRITICAL state is not sufficient to activate Phase 1G. This prevents short-lived geometric disturbances from always launching the detector. The persistent-CRITICAL path is still important because smooth identity drift can remain geometrically continuous and may never create a strict LOST condition.

This trigger design also preserves an important semantic distinction: **CRITICAL is a live reliability state, whereas LOST is a stronger failure state**. Phase 1G may be invoked by either persistent risk or explicit loss.

3 Implementation Artifacts

The following Phase 1G artifacts were created or used during implementation and evaluation. Long filenames are listed separately from their roles to preserve readability in the archival PDF.

- `src/recovery/annotation_candidate_generator.py`
  Oracle/debug candidate generation from UA-DETRAC annotations. Used only for reference evaluation and debugging.
- `src/recovery/real_candidate_generator.py`
  Geometric plausibility filtering for real detector outputs.
- `src/evaluation/evaluate_real_candidate_recall.py`
  Initial real YOLO candidate-recall evaluation across recorded reappearance frames.
- `src/evaluation/evaluate_real_candidate_filtering.py`
  Evaluation of target retention before and after real-candidate filtering.
- `src/evaluation/sweep_phase1g_detector_settings.py`
  Detector input-size/confidence sweep across the eligible reappearance events.
- `src/evaluation/compare_phase1g_runtime.py`
  Initial runtime comparison between practical detector settings.
- `src/evaluation/benchmark_phase1g_detector.py`
  Robust wall-clock benchmark using CUDA synchronization, warm-up, and repeated measurements.
- `src/evaluation/list_phase1g_triggers.py`
  Extraction of event/tracker runs in which Phase 1F activates Phase 1G.
- `src/evaluation/evaluate_phase1g_triggered_candidates.py`
  Trigger-driven candidate-generation simulation from Phase 1F activation until verified reappearance or sequence end.
- `src/evaluation/summarize_phase1g.py`
  Consolidation of final Phase 1G metrics and interpretation.

- `configs/phase1/phase1g_detector.yaml`
  Frozen real-mode detector and plausibility-filter configuration.

Key result files include:

- `results/phase1/phase1g_real_candidate_recall.csv`
- `results/phase1/phase1g_real_candidate_filtering.csv`
- `results/phase1/phase1g_triggered_candidate_results.csv`
- `results/phase1/phase1g_final_summary.txt`
- `results/phase1/phase1g_final_summary_metrics.csv`

4 Candidate Generation Modes

Two deliberately separated modes were retained.

4.1 Oracle / Debug Mode

UA-DETRAC annotations are used to construct candidate vehicles at a frame and apply the same broad plausibility logic. This mode exists for debugging, decomposition of failures, and controlled evaluation. It is **not deployable detector output** and must not be reported as detector recall.

At the 10 eligible recorded-reappearance events, oracle/debug candidate retention was 10/10. The candidate pool changed from 117 raw annotation-backed candidates to 96 plausible candidates, a 17.9% reduction.

4.2 Real Mode

The actual system uses a road-vehicle detector. YOLO11n with COCO classes car, motorcycle, bus, and truck was selected as the lightweight detector baseline. It proposes boxes only. It does not decide which box is the target.

This separation is important for the later failure taxonomy:

- **F4:** true target returned but was not detected / not included in the candidate set.
- **F5:** true target was detected but later identity reasoning selected another candidate or rejected it.

Phase 1G is mainly responsible for preventing F4; later phases address F5.

5 Detector Environment and Infrastructure

A dedicated `vehicle_detector` environment was created rather than reusing the base environment. The final environment used:

- Python 3.10
- PyTorch 2.5.1 + CUDA 12.1 build
- torchvision 0.20.1 + CUDA 12.1 build
- Ultralytics 8.4.173
- CUDA available: true
- GPU: NVIDIA Quadro P2000, compute capability 6.1

YOLO11n was chosen as a lightweight proposal generator, not an identity model. COCO vehicle class IDs were fixed to 2, 3, 5, and 7 (car, motorcycle, bus, truck).

6 Preliminary Detector Sensitivity Check

The first controlled check used E011 at frame 378. This frame contains multiple taxis and therefore provides a useful small-object/similar-distractor stress case. The purpose was to understand how input resolution and detector confidence affect candidate coverage before running all eligible events.

imgsz	conf	Detections	Vehicle recall	Target best IoU	Target found
640	0.25	10	7/10	0.072	No
640	0.15	17	10/10	0.714	Yes
640	0.10	25	10/10	0.714	Yes
960	0.25	11	9/10	0.886	Yes
960	0.15	15	10/10	0.886	Yes
960	0.10	21	10/10	0.886	Yes

Table 1: Initial E011 detector sensitivity study.

This single-frame experiment suggested 960/0.15 as a provisional setting because it achieved complete vehicle recall with fewer detections than 960/0.10. It was explicitly treated as provisional, because one frame is insufficient for freezing a detector configuration.

7 Initial Real Detector Evaluation Across 10 Events

The provisional 960/0.15 setting was then evaluated on the 10 events with recorded reappearance. At each reappearance frame, detector outputs were matched to annotation boxes with an evaluation IoU threshold of 0.30.

Event	GT vehicles	Detections	Frame vehicle recall	Target best IoU	Target proposed
E001	12	29	11/12	0.912	Yes
E002	15	28	13/15	0.801	Yes
E003	11	29	11/11	0.958	Yes
E004	7	30	7/7	0.942	Yes
E005	11	8	5/11	0.810	Yes
E006	10	9	7/10	0.803	Yes
E007	9	14	7/9	0.931	Yes
E008	13	33	13/13	0.941	Yes
E009	19	46	18/19	0.794	Yes
E011	10	15	10/10	0.886	Yes

Table 2: Initial real detector results at 960/0.15. Mean frame-level vehicle recall was approximately 0.866, while true-target candidate recall was 10/10.

This experiment established a key distinction: full-scene vehicle recall does not have to be perfect for Phase 1G to succeed, provided the *selected target* is included in the candidate set. However, the detector generated many proposals (241 across the 10 events), motivating a plausibility filter.

8 Geometric Plausibility Filtering

The filter uses only the initial target reference box and each real detector box; it does not use the true target identity. For candidate box c and reference target box r:

$$\text{area-ratio}(c, r) = \frac{A_c}{A_r}$$

$$\text{aspect-ratio-ratio}(c, r) = \frac{(w_c/h_c)}{(w_r/h_r)}$$

A candidate is retained when:

$$0.35 \leq \text{area-ratio} \leq 15.0$$

and

$$0.5 \leq \text{aspect-ratio-ratio} \leq 2.0.$$

These bounds are intentionally broad. The goal is not to identify the target at this stage, but to remove implausible scale/shape proposals while avoiding premature exclusion of the true target under viewpoint and distance changes.

Under the provisional 960/0.15 configuration, filtering reduced 241 raw detections to 171 candidates (29.0%) while retaining the target in all 10 events. This result was useful but was later superseded for the frozen configuration by the corrected 640/0.20 rerun described below.

9 Detector Setting Sweep

To avoid freezing a detector setting from a single event, five configurations were tested across all 10 eligible reappearance events. Every tested configuration retained the true target in 10/10 events.

imgsz	conf	Target recall	Raw candidates	Filtered candidates
640	0.15	10/10	202	158
640	0.20	10/10	179	146
960	0.15	10/10	241	171
960	0.20	10/10	212	154
960	0.25	10/10	191	143

Table 3: Detector setting sweep across the 10 eligible recorded-reappearance events.

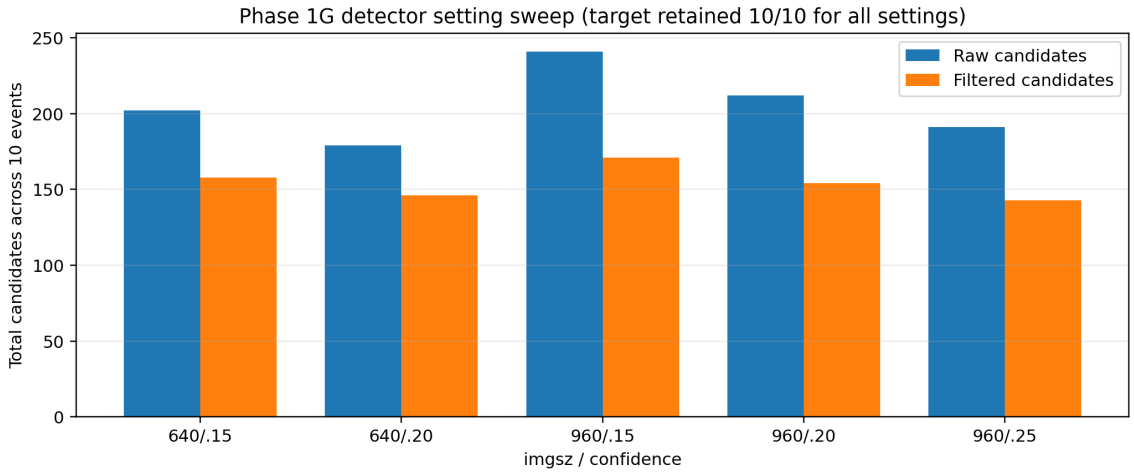


Figure 2: Candidate counts across detector settings. All five settings retained the true target in all 10 tested events.

The two practical candidates were 640/0.20 (146 filtered candidates) and 960/0.25 (143 filtered candidates). The three-candidate difference across all 10 events was too small to justify choosing solely by candidate count, so runtime stability was evaluated.

10 Runtime Comparison and Final Operating Point

An initial Ultralytics-reported timing pass showed similar totals but inconsistent preprocessing/postprocessing behavior. A more robust benchmark therefore used actual wall-clock timing, explicit CUDA synchronization, warm-up, and 50 measurements per setting.

Setting	Mean latency	Median	Std. dev.	Approx. FPS
640 / 0.20	50.18 ms	49.93 ms	0.79 ms	19.93
960 / 0.25	50.12 ms	46.96 ms	20.71 ms	19.95

Table 4: Robust end-to-end detector benchmark.

Mean latency was effectively identical, but 640/0.20 was much more stable. Because it also achieved 10/10 target retention and differed from 960/0.25 by only three filtered candidates across the whole 10-event set, **640/0.20 was frozen as the Phase 1G real-mode operating point**.

11 Frozen Real-Mode Configuration

Parameter	Frozen value
Detector	YOLO11n
Input size	640
Confidence threshold	0.20
Vehicle classes	car, motorcycle, bus, truck
COCO class IDs	2, 3, 5, 7
Device	CUDA device 0
Evaluation target-match IoU	0.30
Minimum area ratio	0.35
Maximum area ratio	15.0
Minimum aspect-ratio ratio	0.5
Maximum aspect-ratio ratio	2.0
Activation trigger	persistent CRITICAL or LOST

Table 5: Frozen Phase 1G real-mode detector and plausibility filter.

The configuration was persisted in `configs/phase1/phase1g_detector.yaml`.

12 Corrected Final Filtering Result at the Frozen Setting

After the operating point was frozen, the real-candidate filtering evaluation was rerun using 640/0.20. This rerun supersedes the earlier 241-to-171 figure from the provisional 960/0.15 setting.

Event	Raw	Filtered	Reduction	Target retained
E001	23	22	4.3%	Yes
E002	21	15	28.6%	Yes
E003	20	19	5.0%	Yes
E004	22	18	18.2%	Yes
E005	7	6	14.3%	Yes
E006	6	6	0.0%	Yes
E007	12	11	8.3%	Yes
E008	26	15	42.3%	Yes
E009	29	24	17.2%	Yes
E011	13	10	23.1%	Yes
Total	**179**	**146**	**18.4%**	**10/10**

Table 6: Final real-mode candidate filtering at the frozen 640/0.20 setting.

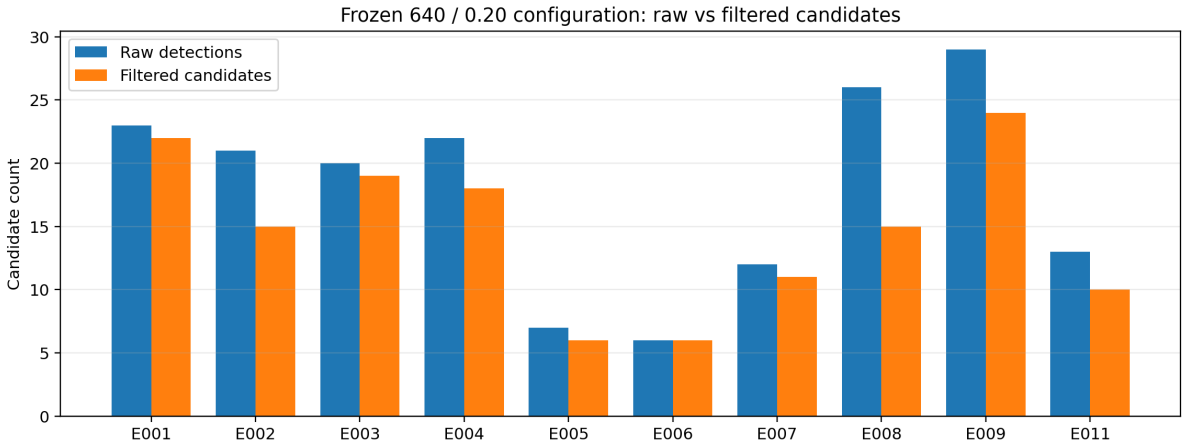


Figure 3: Per-event raw and filtered candidate counts for the final detector configuration.

The final mean filtered candidate count was 14.6 per eligible event. Importantly, the filter reduced the candidate pool without removing the target in this pilot set.

13 Trigger-Driven Integration with Phase 1F

Evaluation at known reappearance frames is useful, but it does not reproduce deployment behavior. Phase 1G was therefore connected to the actual Phase 1F trigger output. Candidate generation begins at the first persistent-CRITICAL or LOST frame and remains active on subsequent frames.

The extracted triggered runs were:

Event	Tracker	Trigger	Persistent CRITICAL	LOST	Verified reappearance
E004	sam21	669	669	–	720
E004	dam4sam	669	669	–	720
E005	sam21	882	–	882	1063
E005	dam4sam	881	–	881	1063
E006	sam21	1094	–	1094	1249
E006	dam4sam	1094	–	1094	1249
E007	sam21	1338	–	1338	1413
E007	dam4sam	1338	–	1338	1413
E012	sam21	591	591	–	–
E012	dam4sam	585	585	604	–

Table 7: Phase 1F tracker-event runs that activated Phase 1G.

This table reveals three qualitatively different trigger types:

- **E004:** false persistent-CRITICAL activation; useful for measuring unnecessary detector cost.
- **E005-E007:** genuine loss/reappearance cases; the main recovery-candidate evaluation.
- **E012:** ambiguity/exit case with no verified reappearance; useful for no-return handling but not recovery success scoring.

14 Trigger-Driven Candidate Evaluation

The detector was run from each Phase 1F trigger frame onward. For events with a verified reappearance, target-proposal scoring was intentionally disabled before that manually verified reappearance frame. This prevents annotation persistence during occlusion from being misinterpreted as visual recovery.

Event	Tracker	Trigger	Reappearance	First target candidate	Detector frames
E004	sam21	669	720	720	52
E004	dam4sam	669	720	720	52
E005	sam21	882	1063	1063	182
E005	dam4sam	881	1063	1063	183
E006	sam21	1094	1249	1249	156
E006	dam4sam	1094	1249	1249	156
E007	sam21	1338	1413	1413	76
E007	dam4sam	1338	1413	1413	76
E012	sam21	591	–	–	18
E012	dam4sam	585	–	–	24

Table 8: Trigger-driven Phase 1G evaluation.

Across the eight triggered runs that had a verified reappearance, the true target was proposed in 8/8. In the six genuine loss/reappearance tracker-runs E005-E007, the true target was proposed in 6/6. The measured proposal delay was 0 frames because the target was already present in the filtered candidate set at the first frame on which recovery scoring became valid.

Interpretation of zero-frame delay: it does not prove that the detector independently discovered the exact visibility-onset frame. The evaluation deliberately begins at the manually verified reappearance frame. Therefore, zero means that *when recovery evaluation first became valid, the target was immediately available in the candidate set.*

15 Final Phase 1G Results

Metric	Final result
Oracle/debug eligible events	10
Oracle/debug true-target retention	10/10
Oracle/debug candidates	117 → 96
Oracle/debug reduction	17.9%
Real-mode eligible reappearance events	10
Raw real detector target recall	10/10
Filtered real detector target recall	10/10
Frozen-setting candidates	179 → 146
Frozen-setting candidate reduction	18.4%
Mean filtered candidates/event	14.6
Total Phase 1F-triggered tracker-runs	10
Triggered runs with verified reappearance	8
Target proposed in eligible triggered runs	8/8
Genuine E005-E007 tracker-runs	6
Genuine E005-E007 successful target proposals	6/6
Mean measured proposal delay	0.00 frames
Total detector frames after triggers	975
False Phase 1F triggers	2
Detector frames caused by false triggers	104
No-reappearance runs	2
Detector frames in no-reappearance runs	42
Robust frozen-detector mean latency	50.18 ms
Robust frozen-detector latency std. dev.	0.79 ms
Approximate frozen-detector throughput	19.93 FPS

Table 9: Final frozen Phase 1G metrics.

16 What Phase 1G Demonstrated

The pilot evidence supports the following claims, stated conservatively:

1. A lightweight real detector can be activated conditionally after Phase 1F reliability failure rather than running as the sole tracking mechanism continuously.

2. At the 10 tested recorded-reappearance frames, the frozen YOLO11n 640/0.20 configuration included the true target in the real candidate set in 10/10 events.

3. Broad geometric plausibility filtering reduced the frozen-setting candidate pool from 179 to 146 detections (18.4%) without removing the true target in the pilot set.

4. In all six genuine E005-E007 loss/reappearance tracker-runs, trigger-driven candidate generation contained the target at the first manually verified reappearance frame.

5. The Phase 1F false-critical behavior has a measurable downstream compute cost: the two E004 false triggers caused 104 detector frames.

6. No-reappearance runs can be kept separate from recovery success/failure: E012 generated 42 detector

frames but was not labeled as either successful or failed recovery because no verified target return exists.

17 Scientific Cautions and Limitations

The following constraints must be preserved in future paper writing.

17.1 Pilot-scale evidence

The 10/10 and 8/8 results are pilot findings from the frozen UA-DETRAC event set, not claims of general detector recall or guaranteed recovery performance on unseen datasets.

17.2 Recorded reappearance versus visibility onset

Recovery scoring starts at manually verified reappearance. Annotation boxes may remain present while a vehicle is largely or fully occluded. Therefore, annotation presence is not treated as equivalent to visual visibility.

17.3 Target recall versus scene recall

The real detector did not achieve perfect full-scene vehicle recall under all settings. Phase 1G's main success criterion is target inclusion in the candidate set, not detection of every road vehicle in the frame.

17.4 Candidate generation is not identity recovery

A candidate set averaging 14.6 filtered vehicles/event is still ambiguous. Phase 1G does not claim to know which vehicle is the original target. The core identity problem remains for the next phases.

17.5 Plausibility filter is intentionally weak

Area and aspect-ratio gates are broad and do not use color, plate, make/model, Re-ID embeddings, or motion identity. Narrower gates might reduce candidate count further but could increase F4 by excluding the correct vehicle after scale/viewpoint change.

17.6 False-trigger overhead matters

Phase 1G cost depends not only on detector speed but also on Phase 1F trigger quality. E004 demonstrates that false reliability alarms can cause substantial unnecessary detector work even when the candidate generator itself is correct.

17.7 Earlier provisional result must not be confused with the frozen result

The 241 → 171 (29.0%) filtering result belongs to the provisional 960/0.15 operating point. The final paper-facing frozen result is **179 → 146 (18.4%) at 640/0.20**. This correction is important because the provisional result appeared in an earlier internal summary before the final configuration was frozen.

18 Failure Separation Enabled by Phase 1G

Phase 1G gives the project a clean experimental boundary for later diagnosis:

- If the returned target is not in the real candidate set, classify the failure as detector/candidate-generation failure (F4).
- If the target is present in the candidate set but later recovery selects another candidate, classify it as identity-selection failure (F5).
- If the target never returns, do not force either F4 or F5; retain the no-reappearance category (F8).

- If multiple candidates remain genuinely indistinguishable, preserve ambiguity rather than forcing a match (F7), to be handled by later decision logic.

This decomposition is essential for the research question because later sequential evidence acquisition can only be assessed fairly if detector availability and identity-decision errors are separated.

19 Phase 1G Freeze Decision

Phase 1G is frozen with the following decisions:

1. Phase 1G is activated by **LOST or persistent CRITICAL** from Phase 1F.
2. The real candidate generator uses **YOLO11n, imgsz=640, confidence=0.20**.
3. Vehicle classes are limited to car, motorcycle, bus, and truck.
4. Real detections are filtered using broad reference-box area and aspect-ratio constraints.
5. Oracle/debug annotation candidates remain available only for evaluation and failure decomposition.
6. Candidate generation is considered successful when the true target is retained; final identity selection is explicitly outside Phase 1G.
7. Trigger-driven evaluation must not count pre-reappearance annotation overlap as recovery.
8. E012-style no-return cases remain unscored for recovery success/failure.

20 Handoff to Phase 1H

The next phase receives a filtered candidate set C_t from Phase 1G. Phase 1H can now evaluate controlled recovery alternatives one component at a time without changing the frozen candidate-generation baseline.

The main next question is no longer *"Can the returned target enter the candidate set?"* Phase 1G has established that baseline for the pilot events. The next question is *"Given multiple plausible vehicles, which identity evidence should be used, in what order, and with what reliability-aware decision rule to MATCH, REJECT, or request more evidence?"*

This is the point at which appearance memory, distractor-aware handling, ambiguity preservation, confidence-dependent routing, and later sequential acquisition become experimentally meaningful.

Archival Note

This document is intended to serve as a source record for later full-paper writing. When results are transferred into the final paper, the frozen metrics and scientific caveats in this document should take precedence over earlier provisional Phase 1G notes. In particular, the paper-facing frozen real-candidate filtering result is 179 raw detections to 146 filtered candidates (18.4%) at YOLO11n 640/0.20 with 10/10 target retention on the 10 recorded-reappearance pilot events.